



## MATHEMATICS METHODS : UNITS 1 & 2, 2020

Test 3 –Trigonometric, Probability, Counting, Exponentials (10%)  
(1.2.10 to 1.2.16, 1.3.6 to 1.3.17, 2.1.1 to 2.1.7)

**Calculator Free - Allow 1 Minute of Reading Time**

| Time Allowed<br>20 Minutes | First Name | Surname | Marks<br>22 marks |
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|----------------------------|------------|---------|-------------------|

Circle your Teacher's Name:

Bestall

Goh

Fraser-Jones

Freer

Koulianos

Luzuk

Rudland

Tanday

**Assessment Conditions:** (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

**1. (3 marks)**

Expand  $(x^2 - 2)^4$  leaving your answer in simplified form.

**2. (4 marks)**

Solve for  $x$ .

$$2 \times 2^{2x} - 16 = +4(2^x)$$

**3. (3 marks)**

Simplify the following expression.

$$\frac{3^{2n} + 9^{n+1}}{9 \times 9^{n+1}}$$

**4. (3, 2, 3 = 8 marks)**

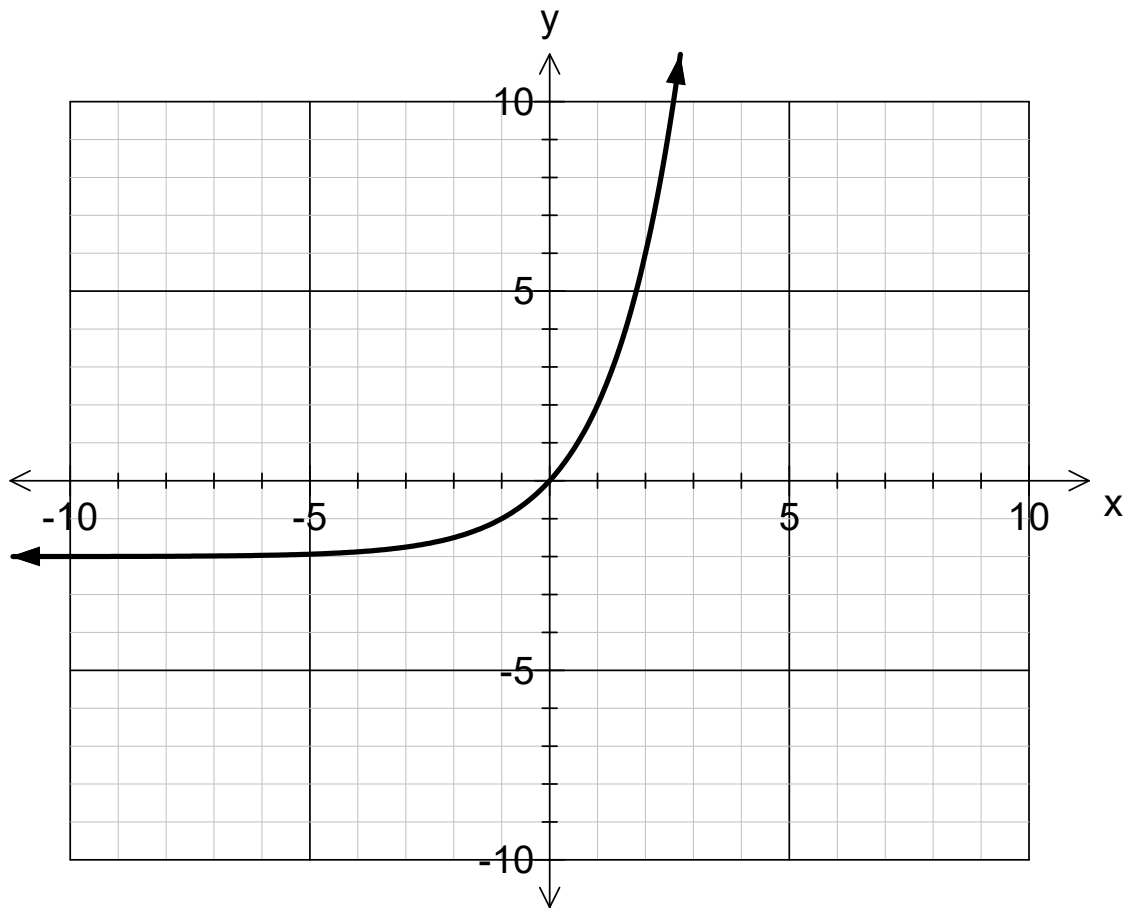
Solve and prove the follow trigonometric functions.

- a. Given the  $A$  and  $B$  are obtuse with  $\sin A = \frac{3}{5}$  and  $\sin B = \frac{5}{13}$ , determine  $\sin(A - B)$ .

**b.** Show that  $\cos\left(x + \frac{\pi}{2}\right) = -\sin(x)$

**c.** Solve  $4 - 4\cos^2 x = 3$  for  $-180^\circ \leq x \leq 90^\circ$

5. (2, 2 = 4 marks)



The exponential function above is a translation of  $y = 2^x$ .

d. Determine the equation for the function.

e. Graph the function  $y = 2^x$  and estimate where the two functions intersect.